

Lab 1 – Product Description

GradPace – Team Copper

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1 Introduction

Workforce expectations are rapidly evolving, and things are honestly getting stressful for people our age. According to recent data, 63% of employers cite skill gaps as their top barrier to growth, while 66% of enterprises are reducing entry-level hiring because of AI. With 91% of reports showing automation is changing or eliminating jobs, finding a decent start after college feels harder than ever. This disconnect is a massive problem across higher education, career services, and the entry-level job market—especially since 48% of recent graduates already feel underqualified for entry-level roles, and 50% to 60% end up underemployed [Cengage Group, 2025].

Right now, schools track our academic progress just fine, but actual career preparation is super fragmented. The tools we have only address isolated needs like coursework tracking, basic job boards, or networking, leaving a huge gap where there's no unified, real-time career readiness roadmap. We end up dealing with fragmented data, delayed gap detection, completely reactive planning, and really limited guidance. That is exactly why we need GradPace to fix this mess and give us a real shot.

2 Product Description

GradPace, a career readiness platform built by Team Copper for students, alumni, and industry professionals that focuses on proactive preparation instead of just throwing traditional job boards at us. Basically, we input our academic and professional info, and the system generates a

personalized roadmap that aligns our current progress with recommendations based on real industry standards. It has some killer core capabilities like early warning gap detection and industry readiness benchmarking, helping identify what we need before entering the job market by integrating live industry data and even GPS based local networking. Because of this, users aren't just stuck tracking past achievements we're synchronized with modern workforce requirements from day one.

Building on this foundation, the platform aims to revolutionize how students transition from the classroom to the modern workforce by eliminating reactive planning and delayed gap detection. Instead of juggling isolated tools like basic course trackers, generic job boards, and disconnected networking apps, users gain access to a centralized ecosystem that continuously aligns their current coursework and skills with actual industry standards. Ultimately, the objective is to empower every student and recent graduate with actionable insights and local professional connections, ensuring they enter the job market fully prepared, highly competitive, and confident in their career trajectory.

2.1 Key Product Features and Capabilities

GradPace functions as a comprehensive career readiness ecosystem that bridges academic progress and professional employment through secure account management, streamlined subscription controls, and dedicated student input interfaces. What makes the platform truly unique and innovative is its departure from traditional, reactive job boards by integrating live industry data to constantly sync student skill sets with real job skill requirements. Rather than just tracking achievements, the platform accomplishes its mission through automated personalized roadmap and a visual dashboard that maps out a student's custom professional path.

This directly solves the problem of fragmented career planning and delayed gap detection by utilizing specialized features like early warning gap detection to spot missing qualifications and industry readiness benchmarking to measure true workforce preparedness. Furthermore, the platform incorporates GPS tracking for local events and direct alumni connections, ensuring users gain actionable guidance and localized networking opportunities

2.2 Major Components (Hardware/Software)

Operating as a web-based platform, the system requires minimal hardware on the user end, relying only on a standard computer and a reliable internet connection to access cloud services. Following the structural design of the CS 410 Major Functional Component Diagram (MFCD), the software architecture is divided into four core components: the frontend, backend, database, and API integrations. The frontend hosts the user interface for account management, information tracking, and dashboard navigation, while the backend processes system logic such as security, roadmap generation, gap detection, benchmark comparisons, and development recommendations. Structured data, including user profiles, academic and professional records, skills, and opportunity listings is securely stored through a PostgreSQL database layer. Additionally, external API integrations enable the platform to connect with live industry data, location services, and AI systems for continuous roadmap refinement and real-time recommendations.

3 Identification of Case Study

Our primary case study focuses on Alex, a computer science undergraduate in his second semester of college who is unsure how to begin professional development outside of general

classwork. Using GradPace opportunity finding tools and custom roadmap features, Alex inputs his transcripts, current work interests, and location information to generate a roadmap that helps him build the specific skills needed for his after graduation goals. This prototype is being built directly for students like Alex who need proactive guidance instead of reactive job hunting, which really gets overwhelming when you are just starting. Looking ahead, GradPace platform could easily expand for future use in full university integrations, deeper alumni engagement, and leveraging aggregated market trend data to help academic institutions keep their actual courses relevant to modern industry demands today.

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Table 1

Table of Comparison Between RWP and Prototype

Feature	Prototype	Real World Product
Account Management	PC display	Live user database
Personal data input	Form entry	Student profile

roadmap	4-year plan but can depend on students overall goals/plan	AI driven personalized paths
Gap Detection	checklist	Real time Alerts & Benchmarking
Local networking events	Simulated event list	GPS-based events

This Table breakdown highlights the smart technical shortcuts GradPace is taking for the prototype without sacrificing the long-term scalability of the platform. By proving out the core mechanics through local simulations and basic interfaces now, a solid groundwork for deploying full AI engines, live databases, and true GPS functionality eventually.

3.1 Prototype Development Challenges

Knowledge Gaps may have API integration issues. GradPace may have issues pulling live data for our industry maps and getting live information from company to assist for our roadmaps.

Database Migration, making sure quires are working correctly for custom roadmaps and mailing sure the data flows. Real-Time data parsing takes actual logic to adjust the students milestone checklist. The designing process may cause some havoc because GradPace is supposed to be a simple design, but we must find a way to grab the user's attention as well with sleek styling.

Other challenges may occur , but I know GradPace will be a good developing tool.

4 Glossary

API (Application Programming Interface): A set of rules that allow different software applications to communicate and exchange data with each other.

Authentication: The process of verifying a user's identity before allowing access to a system.

Backend: The server-side part of a website or application that handles logic, databases, and data processing.

Cloud Computing: The use of remote servers on the internet to store data and run applications instead of a local computer.

Data Leak: The accidental or unauthorized exposure of sensitive or private information.

Database: An organized collection of data that can be stored, accessed, and managed electronically.

Encryption: A security method that converts data into a coded form so that only authorized parties can read it.

Frontend: The user-facing part of a website or application that users interact with in their browser.

Token (AI): A small unit of text processed by an AI model, such as a word, part of a word, or punctuation.

User Interface (UI): The visual elements of a software application that allow users to interact with it.

Web Server: A computer system or software that stores websites and delivers web pages to users when they request them through a browser.

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